# Tour de Maths 2022

## Leg 1

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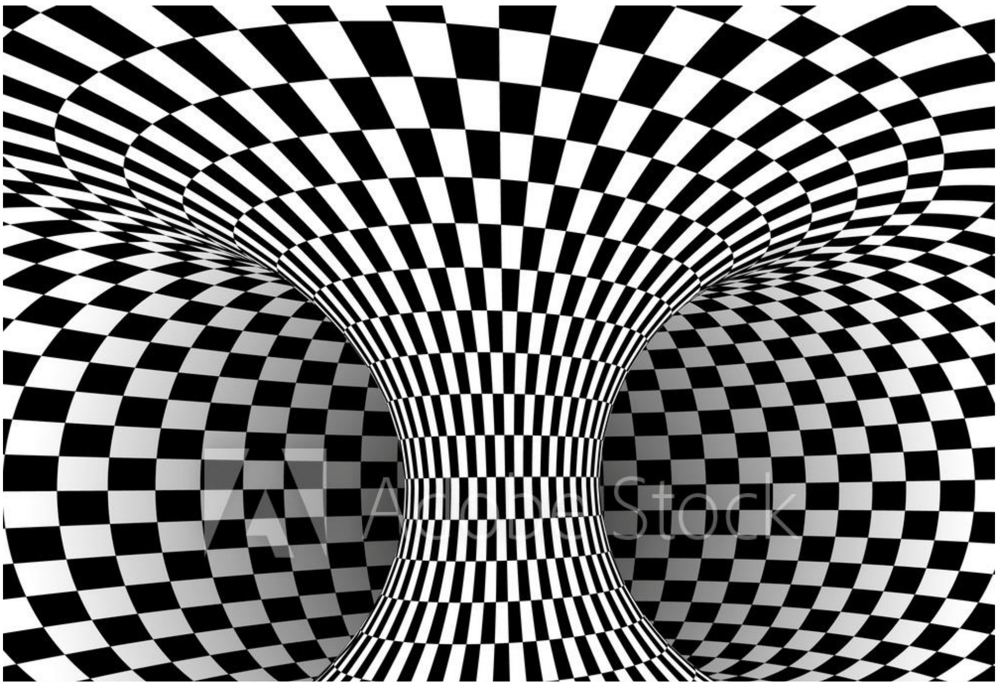
## *Question Paper- MEMO*



# 5 Mark Questions

**Question 1**

| | | | | |
|---|---|---|---|---|
| **M** | **A** | **T** | **H** | **S** |
| **H** | **S** | **M** | **A** | **T** |
| **A** | **T** | **H** | **S** | **M** |
| **S** | **M** | **A** | **T** | **H** |
| **T** | **H** | **S** | **M** | **A** |

The diagram shows a 5 by 5 table.

The top row contains the letters M, A, T, H and S.

The fourth row contains the letters M, A and T as shown.

The remaining squares must be filled with the letters M, A, T, H and S such that no row, column or diagonal contains the same letter more than once.

Complete the table.

**MEMO**

*The* **MAT** *in the* $4^{th}$ *row gives us a clue on how to proceed.*

*Complete* **MATHS** *in that row, by continuing along the row and returning to the beginning of the row when we come to the end of that row.*

*Now do the same in other rows, but make sure that letters are not repeated in the diagonals as well.*

**M** *in* $2^{nd}$ *row will be in* $3^{rd}$ *column. For the* $3^{rd}$ *row, one diagonal has* **M, S** *and* **T** *and other has* **M, A** *and* **S.** *So the middle square of the grid has to be* **H.**

*The last row is now obvious.*

**ANSWER: See diagram**

**Question 2**

How many zeros are there in $3001^2$ , when written in full?

**MEMO**

$(3000 + 1)^2$

$= (3000)^2 + 2(3000) + (1)^2$

$= 9\,000\,000 + 6\,000 + 1$

$= 9\,006\,001$

**ANSWER: 4 zeros**

**Question 3**

| | | | |
|---|---|---|---|
| A | B | C | 17 |
| D | E | F | 14 |
| G | H | I | 15 |
| 17 | P | Q | |

The letters A to I represent numbers.
The numbers are added vertically and horizontally to give the numbers in the last row (17; P; Q) and the last column (17; 14; 15).
Calculate P+Q.

**MEMO**

*The sum of the numbers from A to I is*

17+14+15 = 17+P+Q

$\therefore$ P + Q = 29

**ANSWER: 29**

**Question 4**

The sequence 123456789123456789123… is continued until the sum of all the digits used is 460. The last digit in the sequence is …

(A) 3 (B) 4 (C) 5 (D) 6 (E) 7

**MEMO**

1+2+3+4+5+6+7+8+9 = 45

10 x 45 = 450

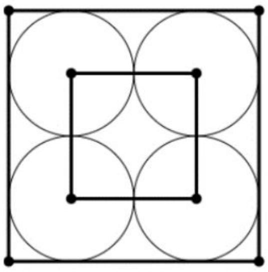
450 + 10 = 460

1+2+3+4 =10

Last digit is 4.

**ANSWER: (B) 4**

**Question 5**



In the diagram, the congruent circles are tangent to the large square and each other as shown; and their centres are the vertices of the small square.

The area of the small square is 4.

Find the area of the large square.

**MEMO**

*Each side of small square has length* 2.

*Radius of each circle is* 1 unit.

*A side of the large square has length*

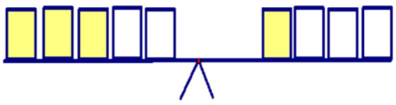
2 x diameter = 4.

*Its* area = 16 sq. units

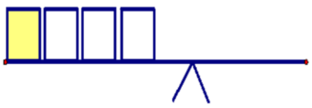
**ANSWER: 16**

**Question 6**

The diagram shows a perfectly balanced scale:



How many shaded rectangles should be placed on the right-hand side of the scale shown below to make it perfectly balanced?



**MEMO**

*Remove 2 shaded and 1 unshaded rectangle respectively from each "side" of scale to conclude that ONE unshaded rectangle weighs the same as TWO shaded rectangles.*

*The 3 unshaded rectangles together with the shaded rectangle weighs the same as 6+1=7 shaded rectangles.*

**ANSWER: 7**

**Question 7**

I spent $\frac{3}{4}$ of my money, then $\frac{2}{5}$ of the rest and I still have R30 left.
How much did I spend?

**MEMO**

Amount : R$x$

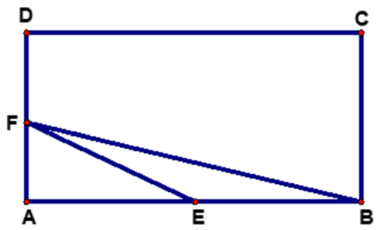
Spent: $\frac{3}{4}x$ Remaining: $\frac{1}{4}x$

Spent: $\frac{2}{5}\left(\frac{1}{4}x\right)$ Remaining: $\frac{3}{5}\left(\frac{1}{4}x\right)$

Now: $\frac{3}{20}x = 30 \therefore x = 200$

Spent: R200 – R30 = R170

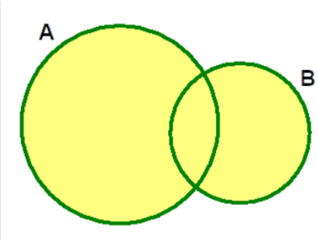
**ANSWER: R170**

| Question | MEMO |
|---|---|
| **Question 8**<br><br><br>ABCD is a rectangle.<br>E and F are the midpoints of AB and AD in that order.<br>If the area of triangle BEF is $10cm^2$,<br>calculate the area of rectangle ABCD, in square centimetres. | **MEMO**<br>Area $\Delta BEF = \frac{1}{2}.EB.FA$<br>$10 = \frac{1}{2}\left(\frac{1}{2}AB\right)\left(\frac{1}{2}AD\right)$<br>$10 = \frac{1}{8}\ AB.AD$<br>Hence $AB.AD = 80$<br>Area of rectangle ABCD $= 80cm^2$<br>**ANSWER: 80$cm^2$** |
| **Question 9**<br>I wrote down the integers 25, 26 27, …, 208.<br>How many digits did I write down? | **MEMO**<br>*From* 25 *to* 99, *there are* 99-25+1=75 *numbers and* 75 x 2 = 150 *digits.*<br>*From* 100 *to* 208 *there are* 208-100+1=109 *numbers and therefore* 109 x 3 = 327 *digits.*<br>*Altogether*: 327 + 150 = 477 *digits.*<br>**ANSWER: 477** |
| **Question 10**<br>In music a demisemiquaver is half of half of half a crotchet, and there are four crotchets in a semibreve.<br>How many demisemiquavers are there in a semibreve? | **MEMO**<br>DSQ = $\frac{1}{8}$ C and 4 C = 1 SB<br>∴1 SB = 4 (8 DSQ)<br>= 32 DSQ<br>**ANSWER:32** |

# 10 Mark Questions

| Question | MEMO |
|---|---|
| **Question 11**<br>A palindromic number is one which reads the same when its digits are reversed, for example 15351.<br>What is the largest six-digit palindromic number which is exactly divisible by 15?<br>-<br>***9+8=17**<br>**9+7=16**<br>**9+6=15**<br>**9+5=14 etc.** | **MEMO**<br>Let the digits of the number be a b c c b a.<br>Last digit must be 0 or 5 to be divisible by 15, but 5 >0. ∴ a = 5<br>5 b c c b 5<br>The sum of the digits must be a multiple of 3.<br>2b + 2c +10 = 3k<br>2(b+c) +10 = 3k<br>2(9+7) +10 =42 which is a multiple of 3<br>**∴b=9 and c=7**<br>**ANSWER: 597795** |

**Question 12**




The diagram shows two overlapping circles, A and B.

The area of the shaded region is $65m^2$ while the overlapping area is $15m^2$.

The ratio of the area of circle A to the area of circle B is 3:2.

By how many *square metres* is circle A larger than circle B?

**MEMO**

*When adding areas of A and B, we are counting the overlapping region twice.*

Area of A + Area of B = 65 +15 = 80 $m^2$

A : B = 3 : 2

Area of A = $\frac{3}{5}(80) = 48$

Area of B = $\frac{2}{5}(80) = 32$

$48 - 32 = 16$

(Or A exceeds B by $\frac{1}{5}$ $(80) = 16)$

**ANSWER: A is 16 $m^2$ larger**

**Question 13**

Two empty containers P and Q have the same volume.
Water flows into P at the rate of 4 litres per minute and into Q at the rate of 6 litres per minute.
After a certain time, container P can still take another 60 litres, but Q has overflowed by 10 litres.
What is the volume of each container?

**MEMO**

*Method 1*:

60+10=70 ; 6-4=2

Water must have been flowing for 35 min.

Volume of tank = 35 x 4 +60 = 200

Or 35 x 6 -10 = 200

*Method 2*:

Let V= volume of each container in litres.

After $t$ minutes, volume of water in P and Q respectively is $4t$ and $6t$.

V -$4t$ = 60 and $6t$ -V =10

Adding the above:

$$2t = 70 \quad then\ t = 35$$

V = 60 + 4(35) = 200

**ANSWER: 200 $l$**

**Question 14**

Sheba and Chante must dust all the ornaments (100 of them!) in their house every week.
If Sheba does it on her own, it takes her 1 hour and 40 minutes.
If Chante does it on her own, it takes her 3 hours and 20 minutes.
If they do it together how long will it take them?

**MEMO**

Sheba: 100 min to dust 100 ornaments i.e. 1 per min.

Chante: 200 min to dust 100 i.e. $\frac{1}{2}$ per min.

Together: $1\frac{1}{2}$ ornaments in 1 min.

$\frac{100}{1,5} = 66,\dot{6}$

100 ornaments in 66 min and 40 sec.

**ANSWER: 66 min 40 sec**

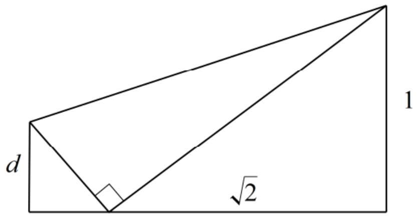
**1h 06min 40sec(66,$\dot{6}$min)**

**Question 15**

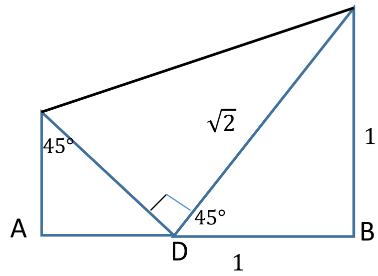
A rectangular sheet of paper with sides $\sqrt{2}$ and $1$ has been folded as shown, so that one corner meets the opposite long edge.

The length $d$ is …

| | | | | |
|---|---|---|---|---|
| (A) $\sqrt{2}-1$ | (B) $\sqrt{2}+1$ | (C) $2\sqrt{2}-2$ | (D) $\sqrt{3}-1$ | (E) $\sqrt{5}-2$ |




**MEMO**




**AD=AB – DB =** $\sqrt{2}-1$ **DB =** $\sqrt{\left(\sqrt{2}\right)^2-1^2}$

(Pythagoras)

**d = AD (sides opp. equal angles)**

**ANSWER: (A)**$\sqrt{2}-1$

# 20 Mark Questions

**Question 16**

Avery and Bryan have the following conversation:

Avery: I am thinking of two different one-digit numbers. Can you guess their sum?

Bryan: No. Can you give me a clue?

Avery: The last digit of their product is the same as your house number.

Bryan: Let me work it out … Now I know!

What is the sum of the two numbers?

*Upper half = lower half*

*Consider number above diagonal.*

*Two different 1 digit numbers – eliminate diagonal.*

*Eliminate all numbers with same last digits.*

**MEMO**

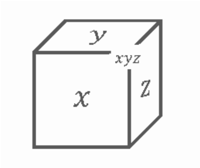
| x | **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** | **9** |
|---|---|---|---|---|---|---|---|---|---|
| **1** | ~~1~~ | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| **2** | 2 | ~~4~~ | 6 | 8 | 10 | 12 | 14 | 16 | 18 |
| **3** | 3 | 6 | ~~9~~ | 12 | 15 | 18 | 21 | 24 | 27 |
| **4** | 4 | 8 | 12 | ~~16~~ | 20 | 24 | 28 | 32 | 36 |
| **5** | 5 | 10 | 15 | 20 | ~~25~~ | 30 | 35 | 40 | 45 |
| **6** | 6 | 12 | 18 | 24 | 30 | ~~36~~ | 42 | 48 | 54 |
| **7** | 7 | 14 | 21 | 28 | 35 | 42 | ~~49~~ | 56 | 63 |
| **8** | 8 | 16 | 24 | 32 | 40 | 48 | 56 | ~~64~~ | 72 |
| **9** | 9 | 18 | 27 | 36 | 45 | 54 | 63 | 72 | ~~81~~ |

*Left with* 21 or 9.

Sum of digits = 7+3=10 or 9+1=10

**ANSWER: 10**

**Question 17**




A mathematician writes numbers on a cube. On each face she writes a positive integer (whole numbers greater than 0). On each vertex she writes the product of its three adjacent faces.For example in the upper right corner she would write the number that's equal to the product of $x, y, z$ .The mathematician does this for each of the vertices.

If the sum of all vertices is 154, what is the sum of all faces?

**MEMO**

Faces: a b c d e f

Vertices: abe+abf+ade+adf + cde+cdf+cbe+cbf

= a(be+bf+de+df) + c(de+df+be+bf)

= (a+c) (be+bf+de+df)

=(a+c)[b(e+f) +d(e+f)]

= (a+c)(b+d)(e+f)

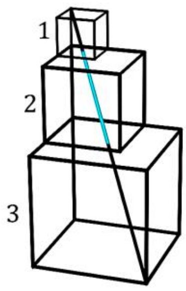
Factors of 154 : 2 x 7 x 11

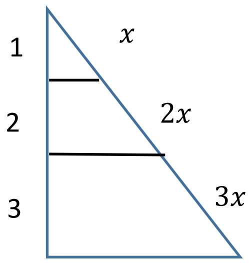
Must be ≥ 2 and factor of 154

(a+c) +(b+d) + (e+f) = 2+7+11=29

**ANSWER: 20**

**Question 18**

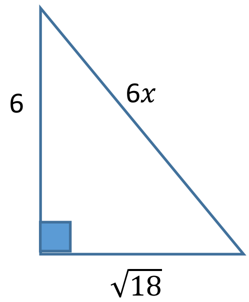


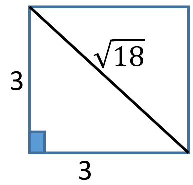




A cube with edge length of 3 is stacked with a cube of edge length 2 and another cube with edge length of 1 as shown, where two of their faces are lined. Imagine drawing a line between opposite corners of the diagram.

What is the length of the line contained inside the middle cube?

**MEMO**




$(6x)^2 = (\sqrt{18})^2 + (6)^2$ **Pythagoras**

$36x^2 = 18 + 36$

$x^2 = \frac{54}{36}$

$x = \sqrt{\frac{54}{36}} = \frac{3\sqrt{6}}{6}$

$x = \frac{\sqrt{6}}{2}$

$2x = \sqrt{6}$

**ANSWER: $\sqrt{6}$**

**Question 19**

Helen and Ivan are carrying the same number of coins. Each coin is either a 20-cent coin or a 50-cent coin.
Helen has a total of 64 of the 20-cent coins and Ivan has a total of 104 of the 20-cent coins.
We don't know how many 50-cent coins either have.

| | Helen | Ivan |
|---|---|---|
| 20 | 64 | 104 |
| 50 | ? | ? |

Who has more money and by how much?

**MEMO**

*Ivan has 40 more 20c coins than Helen.*

*Helen has therefore 40 50c coins more than Ivan.*

Helen – Ivan

-40(0,2) = -8 -8+20=12

+40(0,5) = 20

Helen has R12 more than Ivan.

**ANSWER: Helen has R12 more than Ivan**

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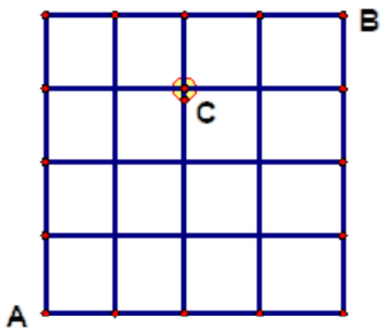
Question 20<br>


<br>
The diagram shows a rectangular network of paths.<br><br>
How many ways are there to move from point A to point B if we can only move up or right without moving through intersection C?
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MEMO<br>
Label each intersection point by the number of possible paths that lead to it from A.<br><br>
Notice that each label is equal to the sum of its two adjacent labels, one on the left and the other below it.<br><br>
Proceeding in this way, and taking care to avoid C, we obtain 40 paths to B.<br><br>
ANSWER: 40
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**Total:200**